

Setting Up LAMP on Amazon Linux with Apache HTTPD

Introduction

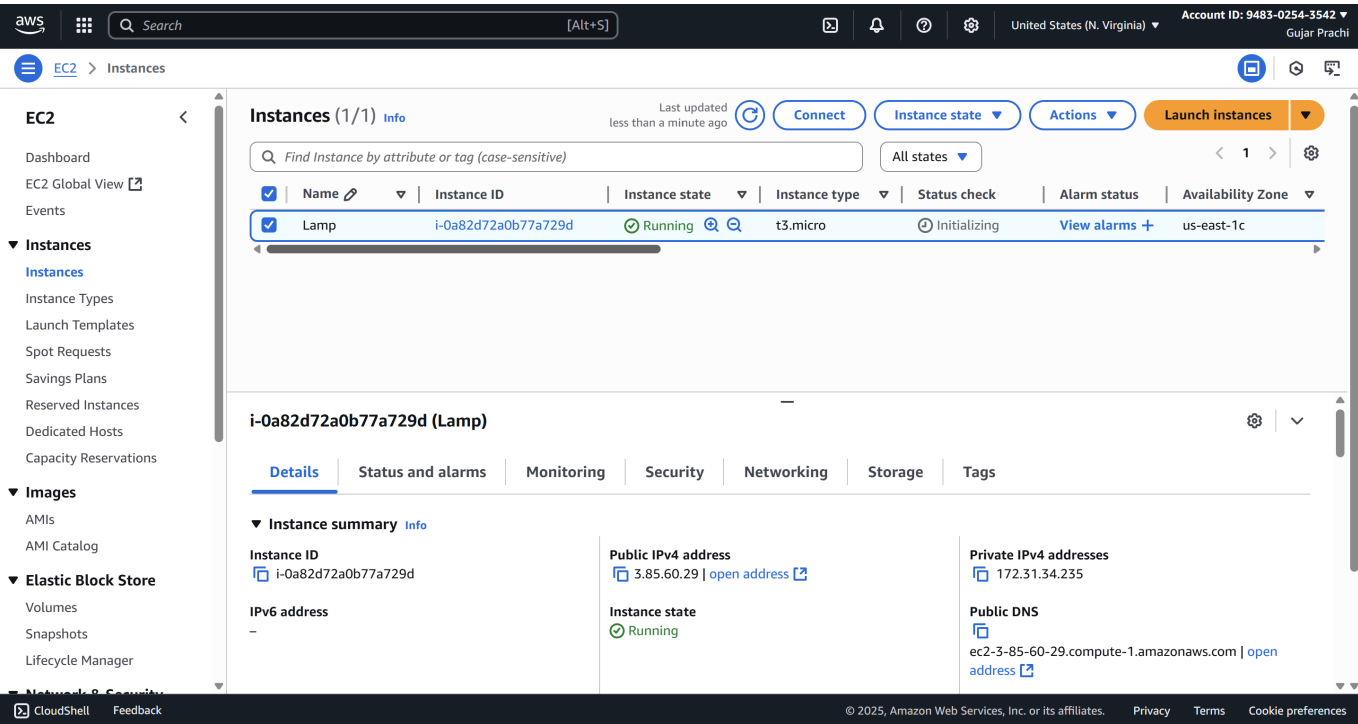
This project explains how to set up a LAMP stack (Linux, Apache, MySQL, PHP) on an Amazon Linux EC2 instance. It covers installing and configuring the Apache HTTPD server, setting up MySQL for database management, and enabling PHP support to serve dynamic content. This setup provides a complete environment for hosting and running web applications on AWS.

Prerequisites

- AWS account (EC2 access)
- Basic Linux knowledge
- SSH key pair
- Security group: ports 80 (HTTP), 22 (SSH)
- Amazon Linux EC2 instance
- Basic knowledge of Apache, MySQL, PHP

Steps to Deploy LAMP Stack

Step 1: Login into AWS Account and Launch Ec2 Instance



Step 2: Establishing a Secure SSH Connection to Your EC2 Instance

- Copy the SSH connection command from the SSH client and paste it into Git Bash to connect to the instance.

Step 3: Update the system and install the Apache HTTPD server.

```
# to install httpd
sudo yum install httpd -y
```

- Start, enable, and check the service status.

```
# to start
sudo systemctl start httpd
```

```
# to enable
sudo systemctl enable httpd
```

```
# to check status
sudo systemctl status httpd
```

```
ec2-user@ip-172-31-34-235:~
[ec2-user@ip-172-31-34-235 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-34-235 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-34-235 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2025-08-23 18:23:26 UTC; 55s ago
     Docs: man:httpd.service(8)
  Main PID: 25493 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 1057)
    Memory: 13.4M
       CPU: 104ms
    CGroup: /system.slice/httpd.service
            └─25493 /usr/sbin/httpd -DFOREGROUND
              └─25494 /usr/sbin/httpd -DFOREGROUND
                └─25495 /usr/sbin/httpd -DFOREGROUND
                  └─25496 /usr/sbin/httpd -DFOREGROUND
                    └─25497 /usr/sbin/httpd -DFOREGROUND

Aug 23 18:23:26 ip-172-31-34-235.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Aug 23 18:23:26 ip-172-31-34-235.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Aug 23 18:23:26 ip-172-31-34-235.ec2.internal httpd[25493]: server configured, listening on: port 80
[ec2-user@ip-172-31-34-235 ~]$
```

Step 4: Install MySQL.

```
# to search for the MySQL package to install
sudo yum search mysql
```

```
# to install.
sudo yum install mariadb105-server -y
```

```

ec2-user@ip-172-31-34-235:~$ sudo yum search mysql
Last metadata expiration check: 0:04:32 ago on Sat Aug 23 18:21:20 2025.
===== Name & Summary Matched: mysql =====
apr-util-mysql.x86_64 : APR utility library MySQL DBD driver
bind-dl2-mysql.x86_64 : BIND server mysql and mysqldyn DLZ modules
collectd-mysql.x86_64 : MySQL plugin for collectd
dovecot-mysql.x86_64 : MySQL back end for dovecot
freeradius-mysql.x86_64 : MySQL support for freeradius
mysql-selinux.noarch : SELinux policy modules for MySQL and MariaDB packages
opendbx-mysql.x86_64 : MySQL backend - provides mysql support in opendbx
perl-DBD-MariaDB.x86_64 : A MySQL interface for Perl
perl-DateTime-Format-MySQL.noarch : Parse and format MySQL dates and times
php8.1-mysqlnd.x86_64 : A module for PHP 8.1 applications that use MySQL databases
php8.2-mysqlnd.x86_64 : A module for PHP 8.2 applications that use MySQL databases
php8.3-mysqlnd.x86_64 : A module for PHP 8.3 applications that use MySQL databases
php8.4-mysqlnd.x86_64 : A module for PHP 8.4 applications that use MySQL databases
postfix-mysql.x86_64 : Postfix MySQL map support
===== Summary Matched: mysql =====
mariadb1011-client-utils.x86_64 : Non-essential client utilities for MariaDB/MySQL applications
mariadb1011-devel.x86_64 : Files for development of MariaDB/MySQL applications
mariadb1011-server-utils.x86_64 : Non-essential server utilities for MariaDB/MySQL applications
mariadb105-devel.x86_64 : Files for development of MariaDB/MySQL applications
mariadb105-server-utils.x86_64 : Non-essential server utilities for MariaDB/MySQL applications
perl-DBD-MariaDB.x86_64 : MariaDB and MySQL driver for the Perl5 Database Interface (DBI)
[ec2-user@ip-172-31-34-235 ~]$ sudo yum install mariadb105-server
Last metadata expiration check: 0:05:19 ago on Sat Aug 23 18:21:20 2025.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
mariadb105-server                      x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux      10 M
Installing dependencies:
mariadb-connector-c                    x86_64            3.3.10-1.amzn2023.0.1    amazonlinux        211 k
mariadb-connector-c-config             x86_64            3.3.10-1.amzn2023.0.1    amazonlinux         9.9 k
mariadb105                             x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux        1.5 M
mariadb105-common                      x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux         28 k
mariadb105-errmsg                      x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux        212 k
mysql-selinux                          noarch            1.0.4-2.amzn2023.0.3     amazonlinux         36 k
perl-B                                  x86_64            1.80-477.amzn2023.0.7    amazonlinux         177 k
perl-DBD-MariaDB                      x86_64            1.22-1.amzn2023.0.4      amazonlinux        153 k
perl-DBI                              x86_64            1.643-7.amzn2023.0.3     amazonlinux        700 k
perl-data-dumper                      x86_64            2.174-460.amzn2023.0.2   amazonlinux         55 k
perl-file-copy                         noarch            2.34-477.amzn2023.0.7    amazonlinux         20 k
perl-filehandle                       noarch            2.03-477.amzn2023.0.7    amazonlinux         15 k
perl-Math-BigInt                      noarch            1:1.9998.39-2.amzn2023.0.2 amazonlinux        202 k
perl-Math-BigRat                      noarch            0.2624-500.amzn2023.0.2  amazonlinux         42 k
perl-Math-Complex                     noarch            1.59-477.amzn2023.0.7    amazonlinux         46 k
perl-sys-hostname                     x86_64            1.23-477.amzn2023.0.7    amazonlinux         16 k
perl-base                             noarch            2.27-477.amzn2023.0.7    amazonlinux         16 k
Installing weak dependencies:
mariadb105-backup                     x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux        6.0 M
mariadb105-cracklib-password-check    x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux         13 k
mariadb105-gssapi-server              x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux         15 k
mariadb105-server-utils               x86_64            3:10.5.29-1.amzn2023.0.1  amazonlinux        207 k
Transaction Summary
-----
Install 22 Packages

Total download size: 20 M
Installed size: 117 M
Is this ok [y/N]: y

```

- Start, enable, and check the service status.

```
sudo systemctl start mariadb
```

```
sudo systemctl enable mariadb
```

```
sudo systemctl status mariadb
```

```

ec2-user@ip-172-31-34-235:~$ sudo systemctl start mariadb
[ec2-user@ip-172-31-34-235 ~]$ sudo systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[ec2-user@ip-172-31-34-235 ~]$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Sat 2025-08-23 18:28:10 UTC; 33s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Main PID: 27702 (mariabdb)
   Status: "Taking your SQL requests now..."
     Tasks: 14 (limit: 1057)
    Memory: 66.4M
       CPU: 492ms
    CGroup: /system.slice/mariadb.service
            └─27702 /usr/libexec/mariabdb --basedir=/usr

Aug 23 18:28:10 ip-172-31-34-235.ec2.internal mariadb-prepare-db-dir[27658]: The second is mysql@localhost, it has no password either, but
Aug 23 18:28:10 ip-172-31-34-235.ec2.internal mariadb-prepare-db-dir[27658]: you need to be the system 'mysql' user to connect.
Aug 23 18:28:10 ip-172-31-34-235.ec2.internal mariadb-prepare-db-dir[27658]: After connecting you can set the password, if you would need to be
Aug 23 18:28:10 ip-172-31-34-235.ec2.internal mariadb-prepare-db-dir[27658]: able to connect as any of these users with a password and without sudo
Aug 23 18:28:10 ip-172-31-34-235.ec2.internal mariadb-prepare-db-dir[27658]: See the MariaDB Knowledgebase at https://mariadb.com/kb
Aug 23 18:28:10 ip-172-31-34-235.ec2.internal mariadb-prepare-db-dir[27658]: Please report any problems at https://mariadb.org/jira
Aug 23 18:28:10 ip-172-31-34-235.ec2.internal mariadb-prepare-db-dir[27658]: The latest information about MariaDB is available at https://mariadb.org/.
Aug 23 18:28:10 ip-172-31-34-235.ec2.internal mariadb-prepare-db-dir[27658]: Consider joining MariaDB's strong and vibrant community:
Aug 23 18:28:10 ip-172-31-34-235.ec2.internal mariadb-prepare-db-dir[27658]: https://mariadb.org/get-involved/
Aug 23 18:28:10 ip-172-31-34-235.ec2.internal systemd[1]: Started mariadb.service - MariaDB 10.5 database server.
[ec2-user@ip-172-31-34-235 ~]$

```

Step 5: Install PHP (FPM Included by Default with Apache HTTPD)

```
# to check the PHP version
php --version
```

```
# to install
sudo yum install php -y
```

```
ec2-user@ip-172-31-34-235:~$ sudo yum install php -y
Last metadata expiration check: 0:08:21 ago on Sat Aug 23 18:21:20 2025.
Dependencies resolved.
=====================================================================================================================================
 Package                                     Architecture                               Version                               Repository                               Size
=====================================================================================================================================
Installing:
php8.4                                     x86_64                                     8.4.10-1.amzn2023.0.1               amazonlinux                               17 k
Installing dependencies:
libsodium                                 x86_64                                     1.0.19-4.amzn2023                   amazonlinux                               176 k
libxslt                                  x86_64                                     1.1.43-1.amzn2023.0.2               amazonlinux                               183 k
nginxfilesystem                          noarch                                     1:1.28.0-1.amzn2023.0.2             amazonlinux                               9.6 k
php8.4-cli                               x86_64                                     8.4.10-1.amzn2023.0.1               amazonlinux                               3.8 M
php8.4-common                            x86_64                                     8.4.10-1.amzn2023.0.1               amazonlinux                               800 k
php8.4-process                           x86_64                                     8.4.10-1.amzn2023.0.1               amazonlinux                               52 k
php8.4-xml                               x86_64                                     8.4.10-1.amzn2023.0.1               amazonlinux                               966 k
Installing weak dependencies:
php8.4-fpm                               x86_64                                     8.4.10-1.amzn2023.0.1               amazonlinux                               2.0 M
php8.4-mbstring                          x86_64                                     8.4.10-1.amzn2023.0.1               amazonlinux                               541 k
php8.4-opcache                           x86_64                                     8.4.10-1.amzn2023.0.1               amazonlinux                               485 k
php8.4-pdo                               x86_64                                     8.4.10-1.amzn2023.0.1               amazonlinux                               100 k
php8.4-sodium                            x86_64                                     8.4.10-1.amzn2023.0.1               amazonlinux                               48 k
Transaction Summary
-----
Install 13 Packages
Total download size: 9.1 M
Installed size: 41 M
Downloading Packages:
(1/13): nginxfilesystem-1.28.0-1.amzn2023.0.2.noarch.rpm                       246 kB/s | 9.6 kB  00:00
(2/13): libsodium-1.0.19-4.amzn2023.x86_64.rpm                                3.8 MB/s | 176 kB  00:00
(3/13): libxslt-1.1.43-1.amzn2023.0.2.x86_64.rpm                              3.6 MB/s | 183 kB  00:00
(4/13): php8.4-8.4.10-1.amzn2023.0.1.x86_64.rpm                              677 kB/s | 17 kB  00:00
(5/13): php8.4-common-8.4.10-1.amzn2023.0.1.x86_64.rpm                       22 MB/s | 800 kB  00:00
(6/13): php8.4-cli-8.4.10-1.amzn2023.0.1.x86_64.rpm                           62 MB/s | 3.8 MB  00:00
(7/13): php8.4-mbstring-8.4.10-1.amzn2023.0.1.x86_64.rpm                     15 MB/s | 541 kB  00:00
(8/13): php8.4-fpm-8.4.10-1.amzn2023.0.1.x86_64.rpm                          26 MB/s | 2.0 MB  00:00
(9/13): php8.4-opcache-8.4.10-1.amzn2023.0.1.x86_64.rpm                      12 MB/s | 485 kB  00:00
(10/13): php8.4-pdo-8.4.10-1.amzn2023.0.1.x86_64.rpm                         3.1 MB/s | 100 kB  00:00
(11/13): php8.4-process-8.4.10-1.amzn2023.0.1.x86_64.rpm                     2.2 MB/s | 52 kB  00:00
(12/13): php8.4-sodium-8.4.10-1.amzn2023.0.1.x86_64.rpm                      2.0 MB/s | 48 kB  00:00
(13/13): php8.4-xml-8.4.10-1.amzn2023.0.1.x86_64.rpm                         31 MB/s | 966 kB  00:00
Total                                                                           43 MB/s | 9.1 MB  00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : php8.4-common-8.4.10-1.amzn2023.0.1.x86_64                  1/1
  Installing     : php8.4-cli-8.4.10-1.amzn2023.0.1.x86_64                    1/13
  Installing     : php8.4-mbstring-8.4.10-1.amzn2023.0.1.x86_64                2/13
  Installing     : php8.4-fpm-8.4.10-1.amzn2023.0.1.x86_64                    3/13
  Installing     : php8.4-opcache-8.4.10-1.amzn2023.0.1.x86_64                4/13
  Installing     : php8.4-pdo-8.4.10-1.amzn2023.0.1.x86_64                    5/13
  Installing     : php8.4-process-8.4.10-1.amzn2023.0.1.x86_64                6/13
  Running scriptlet: nginxfilesystem-1:1.28.0-1.amzn2023.0.2.noarch            7/13
  Installing     : nginxfilesystem-1:1.28.0-1.amzn2023.0.2.noarch              7/13
  Installing     : php8.4-fpm-8.4.10-1.amzn2023.0.1.x86_64                    8/13
  Running scriptlet: php8.4-fpm-8.4.10-1.amzn2023.0.1.x86_64                  8/13
  Installing     : libxslt-1.1.43-1.amzn2023.0.2.x86_64                      9/13
  Installing     : php8.4-xml-8.4.10-1.amzn2023.0.1.x86_64                    10/13
```

- Start, enable, and check the service status.

```
sudo systemctl start php-fpm
```

```
sudo systemctl enable php-fpm
```

```
sudo systemctl status php-fpm
```

```
ec2-user@ip-172-31-34-235:~  
[ec2-user@ip-172-31-34-235 ~]$ php --version  
PHP 8.4.10 (cli) (built: Jul 2 2025 02:22:42) (NTS gcc x86_64)  
Copyright (c) The PHP Group  
Built by Amazon Linux  
Zend Engine v4.4.10, Copyright (c) Zend Technologies  
with Zend OPcache v8.4.10, Copyright (c), by Zend Technologies  
[ec2-user@ip-172-31-34-235 ~]$ sudo systemctl start php-fpm  
[ec2-user@ip-172-31-34-235 ~]$ sudo systemctl enable php-fpm  
Created symlink /etc/systemd/system/multi-user.target.wants/php-fpm.service → /usr/lib/systemd/system/php-fpm.service.  
[ec2-user@ip-172-31-34-235 ~]$ sudo systemctl status php-fpm  
● php-fpm.service - The PHP FastCGI Process Manager  
   Loaded: loaded (/usr/lib/systemd/system/php-fpm.service; enabled; preset: disabled)  
   Active: active (running) since Sat 2025-08-23 18:31:55 UTC; 29s ago  
     Main PID: 28319 (php-fpm)  
    Status: "Processes active: 0, idle: 5, Requests: 0, slow: 0, Traffic: 0.00req/sec"  
       Tasks: 6 (limit: 1057)  
      Memory: 9.9M  
         CPU: 48ms  
    CGroup: /system.slice/php-fpm.service  
            └─28319 "php-fpm: master process (/etc/php-fpm.conf)"  
              └─28320 "php-fpm: pool www"  
                └─28321 "php-fpm: pool www"  
                  └─28322 "php-fpm: pool www"  
                    └─28323 "php-fpm: pool www"  
                      └─28324 "php-fpm: pool www"  
  
Aug 23 18:31:55 ip-172-31-34-235.ec2.internal systemd[1]: Starting php-fpm.service - The PHP FastCGI Process Manager...  
Aug 23 18:31:55 ip-172-31-34-235.ec2.internal systemd[1]: Started php-fpm.service - The PHP FastCGI Process Manager.  
[ec2-user@ip-172-31-34-235 ~]$ |
```

Step 6: Navigate to Apache HTTPD's default directory and create the files index.html and index.php.

```
# to navigate to default directory  
sudo /var/www/html
```

```
# to create file  
sudo vim file_name
```

```
# to view the code  
cat file_name
```

```
#to restart and to verify the applied changes  
sudo systemctl restart nginx
```

```
ec2-user@ip-172-31-34-235:/var/www/html
[ec2-user@ip-172-31-34-235 ~]$ cd /var/www/html/
[ec2-user@ip-172-31-34-235 html]$ ls
[ec2-user@ip-172-31-34-235 html]$ sudo vim index.html
[ec2-user@ip-172-31-34-235 html]$ cat index.html
<h1> This is my LAMP website </h1>

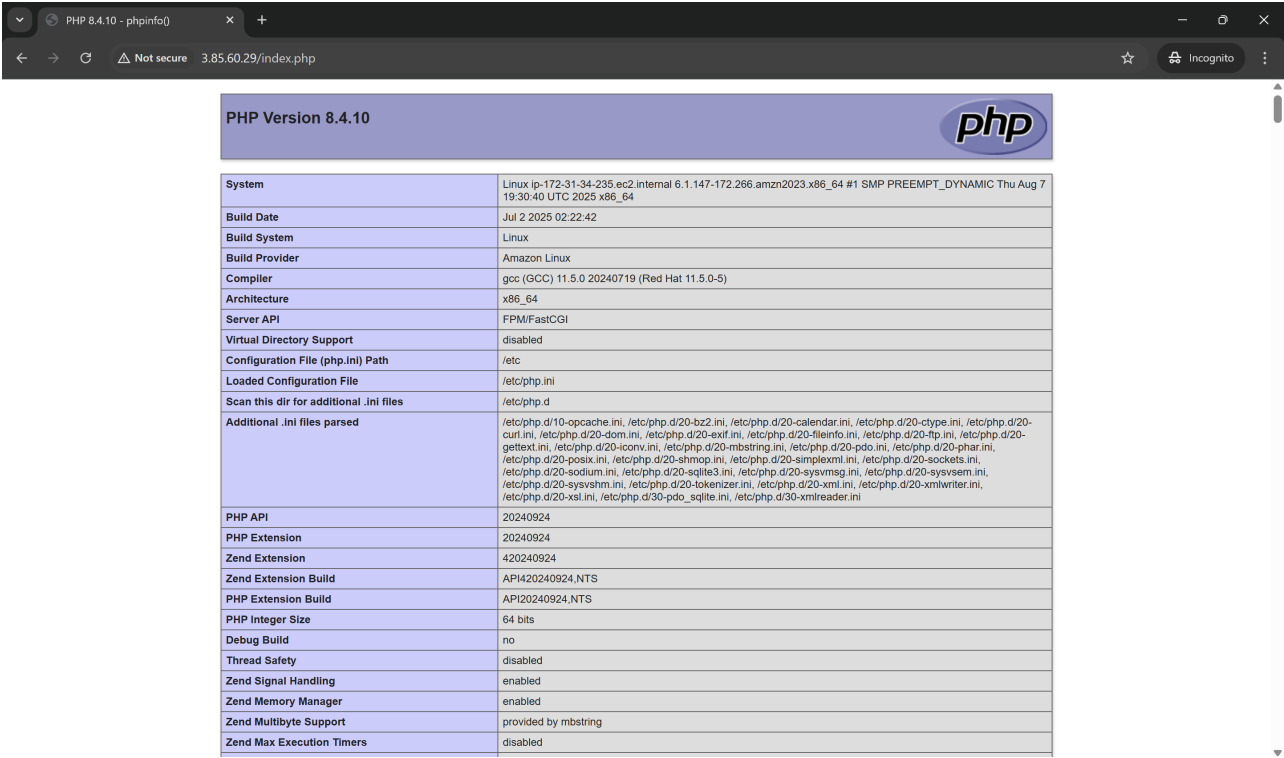
[ec2-user@ip-172-31-34-235 html]$ sudo vim index.php
[ec2-user@ip-172-31-34-235 html]$ cat index.php
<?php
    phpinfo();
?>
[ec2-user@ip-172-31-34-235 html]$ sudo systemctl restart httpd mariadb php-fpm
[ec2-user@ip-172-31-34-235 html]$ |
```

Step 7: Copy the public IP address and paste it into the browser to access the website.

- The index.html file is the default page.



- To view the PHP page, enter publicip/index.php in the URL bar.



Summary

This project demonstrates how to set up a LAMP stack on Amazon Linux using the Apache HTTPD server. It covers connecting to an EC2 instance via SSH, updating and installing required packages, configuring Apache, MySQL, and PHP, and creating sample files to test the setup. Finally, the hosted site is accessed using the instance’s public IP.